# Mr. Ajit Jadhav

## Generative AI Engineer

www.linkedin.com/in/ai-ajitjadhav
ajit.jadhav.ai@outlook.com
+91 9421424227
Pune, India

## Career Objective

To pursue a lead role with AI architecture ownership, from design through deployment, with 4.5 years of experience building production-grade Generative AI and agentic AI systems.

## Core Competencies

**Agentic AI & Harness Engineering:** Agent Harness Engineering (Agent = LLM + Harness) | Agent Loops & Planning | Tool Design & Tool Gateway | Memory & State Persistence | Sandboxed Execution | LLM & Agent Evaluation | LangGraph | Strands Agents | LangChain | MCP | A2A | Agent Skills | Supervisor, Swarm & Planner-Executor Patterns | Human-in-the-Loop | Agent Routing | Ollama

**Generative AI:** Advanced RAG (Hybrid Search, Reranking, GraphRAG) | Vector Databases | Context Engineering | Prompt Caching | Structured Outputs | Multimodal & Voice AI | LiteLLM Model Routing

**ML & Model Engineering:** Transformers & Attention | Tokenization | Embeddings | Fine-Tuning (LoRA, QLoRA, PEFT) | Quantization & Distillation | Model Serving & Inference Optimization (vLLM)

**AWS Cloud & AI Technologies:** Amazon Bedrock | AgentCore | SageMaker AI | Amazon Nova Sonic | Textract | Transcribe | Kendra | OpenSearch Serverless | Lambda | ECS/Fargate | API Gateway | DynamoDB | Cognito | IAM | KMS | WAF | CloudWatch | VPC | Amazon Lex | Amazon Polly | Amazon Nova Act | Kiro

## Professional Experience

### C.G. IT Services LLC | Generative AI Engineer — Oct 2024 - Present

**Project: NEXUS (Next-Generation Experience User System) | Client: Rivian**

- Designed and deployed an enterprise-grade agentic AI platform on AWS Bedrock AgentCore Runtime, leveraging LangGraph orchestration for stateful multi-agent workflows and a LiteLLM-based multi-provider gateway (Anthropic, OpenAI, Meta) with dynamic routing and cost optimization, integrating RAG-based semantic search (OpenSearch Serverless + Titan Embeddings v2 ) for contextual knowledge retrieval.
- implementing real-time guardrails for prompt injection detection and PII protection to ensure secure and reliable AI interactions. Implemented observability using OpenTelemetry and Langfuse, automated infrastructure via Terraform, and optimized performance through intelligent routing, achieving 70% reduction in manual effort, 25% cost savings, 96% retrieval accuracy, and 1000+ daily interactions with sub-500 ms latency in distributed cloud environments.

**Project: SupplyMind (AI-Powered Supply Chain Intelligence Platform) | Client: Cardone Industries**

- Worked on architecting and deploying an end-to-end multi-agent AI orchestration platform on AWS Bedrock for enterprise supply chain operations, implementing five specialized agents with intelligent supervisor routing, using serverless architecture (Lambda, AppSync, DynamoDB, RDS PostgreSQL, S3) for scalable, real-time operations and efficient distributed workload processing.
- Integrated semantic search (OpenSearch Serverless), ML-based forecasting (SageMaker), and event-driven pipelines (Kinesis, Glue, EventBridge, ElastiCache), with end-to-end security (Cognito, IAM, KMS, WAF) and infrastructure automation using AWS CDK and CloudFormation for highly available and fault-tolerant systems.

### C.G. IT Services LLC | Cloud & AI Associate (Contract) — April 2024 - Sep 2024

**Project: Catalyst (enterprise generative AI knowledge platform) | Client: Lucid Motors**

- Deployed and managed an enterprise Generative AI knowledge platform on AWS cloud data centre, unifying datasets and enabling global teams with RAG-powered, accurate knowledge access for scalable enterprise knowledge discovery and intelligent information retrieval systems. Implemented a secure inference workflow using S3, CloudFront, and Cognito for authentication, routing queries via API Gateway and Lambda for preprocessing and embedding generation, and deploying HuggingFace InstructorXL with LLMs (Falcon 7B-Instruct, Llama-2 13B-Chat) on SageMaker integrated with OpenSearch for enterprise grade Q&A and structured conversational systems.
- Migrated data from OCI and GCP to AWS using secure Site-to-Site VPN connectivity, enabling encrypted data transfer and centralized, scalable storage in Amazon's object storage.

### Duroshox Pvt. Ltd. | GET (Infrastructure & Maintenance) — June 2023 - Oct 2023

- Built foundational experience in bare metal and distributed infrastructure, configuring networking hardware and maintaining GPU clusters optimized for AI workloads and high-performance computing environments.

### AllOps Technologies Pvt. Ltd. | Big Data and Cloud Associate — Nov 2021 - May 2023

- Deployed and managed a production-grade Cloudera Hadoop ecosystem and AWS Data Lakehouse, integrating EMR, S3, RDS, MSK, Glue, Redshift, Athena, and QuickSight for real-time IoT analytics.
- Monitored Kafka Connect and EMR pipelines, optimized auto-scaling, and managed S3 datasets to ensure reliable, high-performance data processing for business-critical workloads.

## Certification

**OCI 2025 Certified Generative AI Professional**

## Education

**Bachelor of Engineering | Pune University 2022**